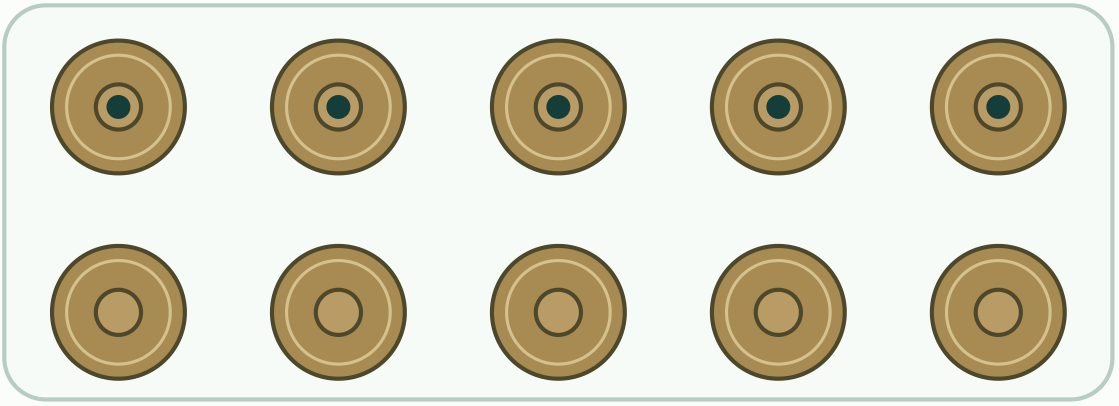
Two ballots. One unseen vote.

Modern ten-person training panel; starting rehearsal: last-vote. These votes are invented.

Open this HTML without a connection. Use the browser's Print command for PDF. Reopen the JSON in Psephos to continue editing.

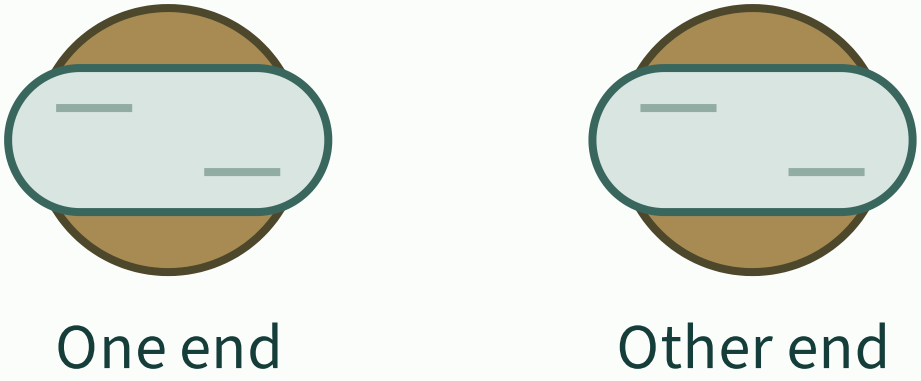
Equal votes. The defendant wins.



5 pierced for the prosecutor; 5 solid for the defendant. ἂν δὲ ἴσαι, ὁ φεύγων: equality favors the defendant.

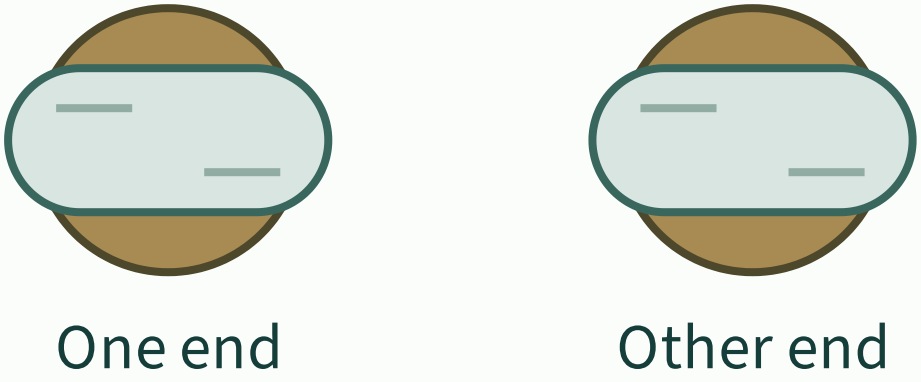
Bronze contains the chosen ballots; wood contains their unused opposites. Only bronze is authoritative under 68.3. The ten positions and vessel drawings are new teaching devices.

Current observer view



Private selection: solid. Both axle ends covered; the selected ballot cannot be distinguished in this drawing.

Pinned observation



Private choice: solid. Both axle ends covered; the selected ballot cannot be distinguished in this drawing.

What the count reveals

This is a modern inference model beside the ancient grip (68.4) and count (69.1). Juror labels and assumed knowledge are new teaching devices.

Public count: 5 pierced, 5 solid. **Target:** Juror 10.

Assumed knowledge: Juror 1: pierced; Juror 2: pierced; Juror 3: pierced; Juror 4: pierced; Juror 5: pierced

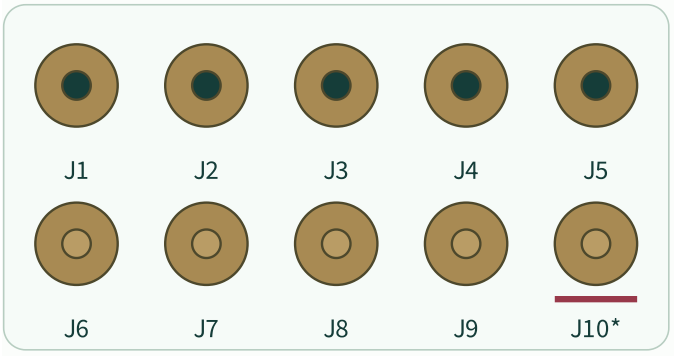
Juror 10's choice is forced: solid.

1 arrangement fits these facts. Every compatible arrangement gives the target the same ballot. The grip can hide a choice during casting without preventing this deduction from the final count and the assumptions.

Choices still possible for Juror 10

Target's ballot	Compatible arrangements
Pierced	0
Solid	1

Compatible arrangement A: target solid



* Target juror. Holes mean pierced; closed centers mean solid.

▶ Why this count is complete

Your explanation

> Five and five acquit under ἂν δὲ ἴσαι. Replacing a solid ballot leaves ten in total but changes the result. <This is my explanation.>

Selected sources: When the votes are equal (69.1, verdict clause). The prose is preserved, not graded.

The rehearsal sequence

Nine invented prior votes: five pierced, four solid.

1. **Refused:** Refuse this pair: each juror must receive one pierced and one solid ballot, shown to both litigants. (issue · source: pair)
2. **Accepted:** One pierced and one solid ballot received. Choose the one that will count. (issue · source: pair)
3. **Accepted:** The pierced ballot is selected for bronze. The other goes into wood. (choose · source: grip)
4. **Accepted:** The axle ends are exposed. The observer can now distinguish the selected ballot. (cover · source: grip)
5. **Accepted:** Both axle ends are covered in the observer drawing. (cover · source: grip)
6. **Accepted:** The solid ballot is selected for bronze. The other goes into wood. (choose · source: grip)
7. **Accepted:** Juror 10: solid into bronze; pierced into wood. (cast · source: grip)
8. **Accepted:** The counted urn is open. Separate pierced from solid ballots. (count · source: urns)
9. **Accepted:** One additional pierced ballot has been inserted in the modern counting exercise. (intervene · source: count)
10. **Accepted:** One solid ballot has been replaced by a pierced ballot in the modern counting exercise. (intervene · source: count)
11. **Accepted:** Original count restored. (intervene · source: count)

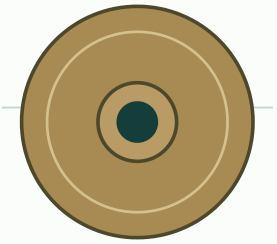
Modern counting interventions

Modern interventions in the bronze count

Count	Pierced	Solid	Total	Ten jurors?
Original	5	5	10	Matches
Extra pierced	6	5	11	Mismatch
Solid replaced	6	4	10	Matches

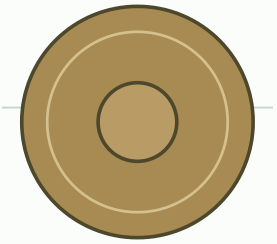
A matching total alone does not establish a ballot's history. Interventions are deliberate modern exercises, outside the described ancient procedure.

The pair and its meaning



Pierced axle

First speaker



Solid axle

Second speaker

Read the procedure

One of each · 68.2, opening

ψῆφοι δέ εἰσιν χαλκαῖ, αὐλίσκον ἔχουσαι ἐν τῷ μέσῳ, αἱ μὲν ἡμίσειαι τετρυπημέναι, αἱ δὲ ἡμίσειαι πλήρεις. οἱ δὲ λαχόντες ἐπὶ τὰς ψήφους, ἐπειδὰν εἰρημένοι ὦσιν οἱ λόγοι, παραδιδόασιν ἑκάστῳ τῶν δικαστῶν δύο ψήφους, τετρυπημένην καὶ πλήρη, φανερὰς ὁρᾶν τοῖς ἀντιδίκοις, ἵνα μήτε πλήρεις μήτε τετρυπημένας ἀμφοτέρας λαμβάνωσι.

The ballots are bronze, with a little tube in the middle: half are pierced, half solid. When the speeches have been delivered, the officials allotted to the ballots hand each juror two ballots, one pierced and one solid, in plain view of the opposing parties, so that jurors receive neither two solid ones nor two pierced ones.

The pair is shown openly before the choice is concealed. This selection ends before the damaged passage about pay-tokens. The editor's sentence punctuation is rendered with a modern full stop.

ψῆφοι (psêphoi)
Ballots; plural of ψῆφος. The word can also mean a pebble used in counting or voting. Here the text explicitly says bronze.

αὐλίσκον (aulískon)
A little tube. Its two ends are the distinguishing feature that the grip conceals.

τετρυπημέναι (tetrupēménai)
Having been pierced: a perfect passive participle describing the resulting physical state. The ballot is not inscribed with a candidate's name.

πλήρεις (plḗreis)
Full or solid. In this context the axle has no through-hole.

φανερὰς (phaneràs)
Visible, in plain view. The parties inspect what is issued; this is a different moment from watching a concealed vote.

ἵνα μήτε (hína mḗte)
So that neither… The purpose clause explains why showing the pair matters.

Only bronze counts · 68.3, final clause

ὁ μὲν χαλκοῦς κύριος, ὁ δὲ ξύλινος ἄκυρος, ἔχων ὁ χαλκοῦς ἐπίθημα διερρινημένον, ὥστ᾽ αὐτὴν μόνην χωρεῖν τὴν ψῆφον, ἵνα μὴ δύο ὁ αὐτὸς [ἐμβάλῃ].

The bronze one is authoritative; the wooden one has no authority. The bronze one has a lid with a filed opening, so that only the ballot itself fits, to prevent the same person from putting in two.

The implied nouns are the two urns named earlier in 68.3. Square brackets preserve Kenyon's editorial supply [ἐμβάλῃ]. The drawing uses schematic vessels: it is not evidence of their precise shape. A narrow opening is a stated precaution, not a proof that fraud was impossible.

χαλκοῦς (khalkoûs)
Bronze. It distinguishes the material of the authoritative urn, not the side supported by a vote.

κύριος (kúrios)
Authoritative or effective. Both pierced and solid ballots can be authoritative when placed in this urn.

ξύλινος (xúlinos)
Wooden. The unchosen ballot goes here whatever the juror's choice was.

ἄκυρος (ákuros)
Without authority, not counting. This does not mean that the ballot itself is broken or malformed.

μόνην (mónēn)
Alone or only. The slot admits the ballot by itself; the text gives this as a precaution against putting in two.

The concealed choice · 68.4, after the evidence challenge

ἔπειτα πάλιν ἀνακηρύττει· ‘ἡ τετρυπημένη τοῦ πρότερον λέγοντος, ἡ δὲ πλήρης τοῦ ὕστερον λέγοντος.’ ὁ δὲ δικαστὴς λα[βὼν ἅμα] ἐκ τοῦ λυχνείου τὰς ψήφους, πιέζων τὸν [αὐλίσκον] τῆς ψήφου καὶ οὐ δεικνύων τοῖς ἀγωνιζομένοις οὔτε τὸ τετρυπημένον οὔτε τὸ πλῆρες, ἐμβάλλει τὴν μὲν κυρίαν εἰς τὸν χαλκοῦν ἀμφορέα, τὴν δὲ ἄκυρον εἰς τὸν ξύλινον.

Then the herald proclaims again: ‘The pierced one for the first speaker, the solid one for the later speaker.’ The juror [takes both ballots together] from the stand, grips the ballot's [little tube], and shows the litigants neither the pierced nor the solid end. He puts the effective ballot into the bronze urn, the ineffective one into the wooden urn.

The herald is the subject supplied from the preceding sentence. λα[βὼν ἅμα] and [αὐλίσκον] retain the printed editor’s supplies. The diagram isolates the ends as seen by a spectator; it does not simulate every sight line, texture, or possible clue. This app records private choices for learning and is not a private voting system.

πρότερον λέγοντος (próteron légontos)
Of the one speaking first. The herald initially defines the choice by speaking order.

ὕστερον λέγοντος (hústeron légontos)
Of the one speaking later: the opposing side. In 69.1 the sides are named as prosecutor and defendant.

πιέζων (piézōn)
Pressing or gripping. This physical act conceals the difference between the axle ends.

οὐ δεικνύων (ou deiknúōn)
Not showing. The same juror's hands conceal both ends, rather than marking a third kind of ballot.

κυρίαν (kurían)
The effective one. The selected ballot is defined by where it is placed; pierced is not inherently the effective kind.

Make the count visible · 69.1, opening

πάντες δ’ ἐπειδὰν ὦσι διε[ψηφισ]μένοι, λαβόντες οἱ ὑπηρέται τὸν ἀμφορέα τὸν κύριον, ἐξερῶσιν ἐπὶ ἄβακα τρυπήματα ἔχοντα ὅσαιπερ εἰσὶν αἱ ψῆφοι, καὶ ταῦτα ὅ[πως] αἱ κυρίαι προκείμεναι εὐαρίθμητοι ὦσιν, καὶ τὰ τρυπητὰ καὶ τὰ πλήρη, [[δηλονότι τοῖς ἀντιδίκοις]].

When everyone has [finished voting], the attendants take the authoritative urn and empty it onto a board with as many holes as there are ballots. This is [so that] the effective ballots, laid out, may be easy to count, both pierced and solid, [[plainly, to the opposing parties]].

The supplies δι[ψηφισ]μένοι and ὅ[πως] follow the print. Kenyon puts the final phrase in double brackets; the apparatus treats it as a gloss to exclude. The English preserves that distinction. The training board's two-by-five layout is new.

πάντες (pántes)
Everyone. This is why the exercise holds back the count while a training juror has yet to vote.

ἄβακα (ábaka)
A board or counting surface; accusative of ἄβαξ. Here its holes make the ballots visible for counting.

τρυπήματα (trupḗmata)
Holes in the board. Do not confuse these with the holes in some ballot axles.

εὐαρίθμητοι (eu aríthmētoi)
Easy to count. The arrangement exposes the aggregate result while the casting grip conceals an individual's choice.

When the votes are equal · 69.1, verdict clause

καὶ ἀναγορεύει ὁ κήρυξ τὸν [ἀριθμὸν] τῶν ψήφων, τοῦ μὲν διώκοντος τὰς τετρυπημένας, τοῦ δὲ φεύγοντος τὰς πλήρεις· ὁποτέρῳ δ' ἂν πλείων γένηται, οὗτος νικᾷ, ἂν δὲ ἴσαι, ὁ φεύγων.

The herald announces the [number] of votes: pierced for the prosecutor, solid for the defendant. Whichever side gets the greater number wins; but if they are equal, the defendant wins.

[ἀριθμὸν] retains the editorial supply. The last clause omits the repeated verb 'wins'. The app gives a verdict only within this described counting procedure. It does not judge the merits of a real dispute or model the separate penalty assessment of 69.2.

διώκοντος (diṓkontos)
Of the pursuer or prosecutor. The participle names a legal role, not a claim that the prosecution is correct.

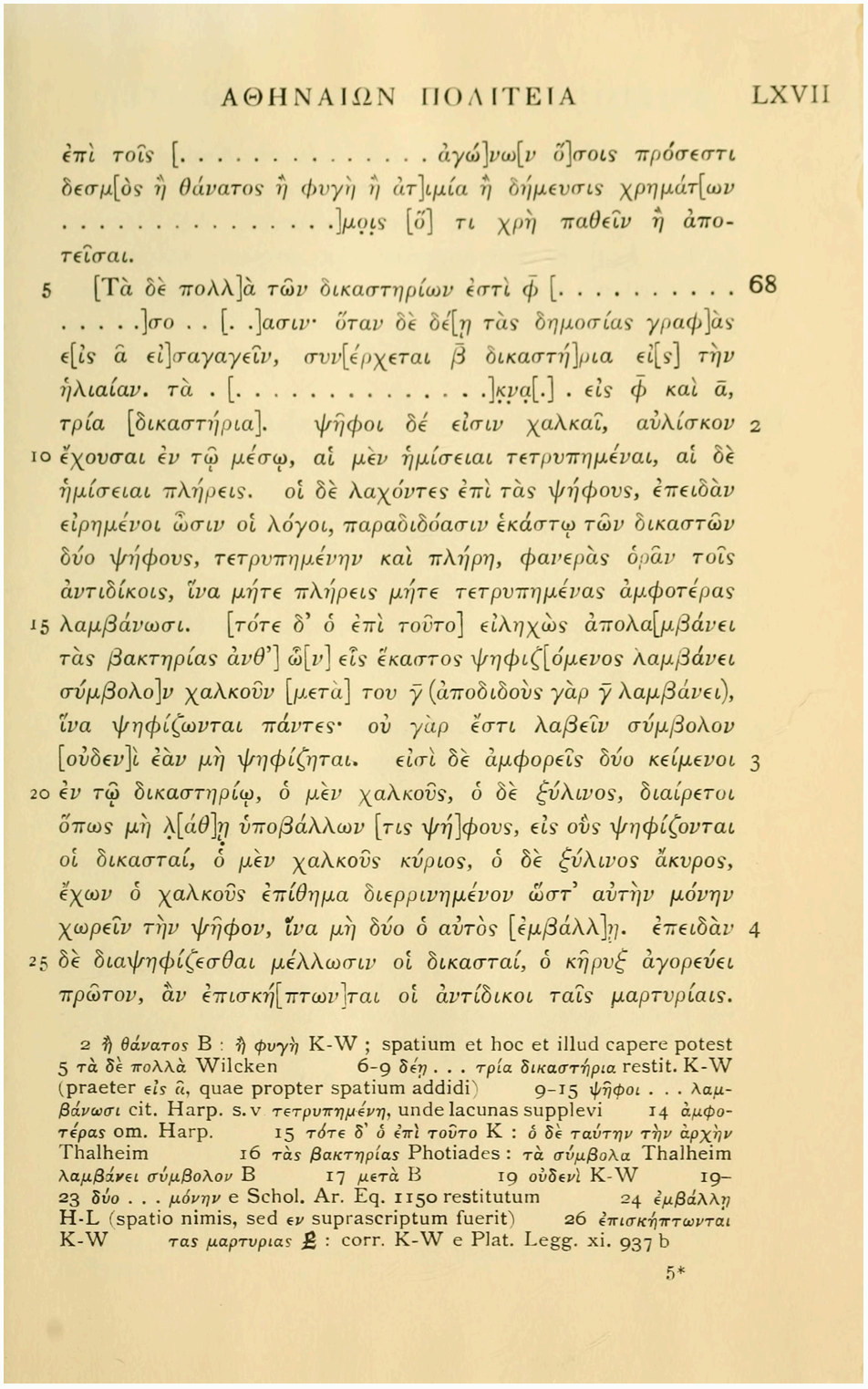
φεύγοντος (pheúgontos)
Of the defendant: literally 'the one fleeing'. Legal context gives this familiar verb its technical role.

πλείων (pleíōn)
Greater in number, referring to ψῆφος or the vote. A prosecution win requires strictly more pierced ballots.

ἂν δὲ ἴσαι (àn dè ísai)
But if [the votes are] equal. ἴσαι is feminine plural, matching the understood ψῆφοι. Equal is not ‘more than half’.

ὁ φεύγων (ho pheúgōn)
The defendant [wins]. The last words settle the five-to-five training count without another ballot.

The complete printed witness



ΑΘΗΝΑΙΩΝ ΠΟΛΙΤΕΙΑ LXVII

ἐπὶ τοῖς [. . . . . ἀγώ]νω[ν ὅ]σοις πρόσεστι
δεσμ[ὸς ἢ θάνατος ἢ φυγὴ ἢ ἀτ]ιμία ἢ δήμευσις χρημάτ[ων
. . . . . .]μ̣οις [ὅ] τι χρὴ παθεῖν ἢ ἀπο-
τεῖσαι.
5 [Τὰ δὲ πολλ]ὰ τῶν δικαστηρίων ἐστὶ φ̄ [. 68
. . . . .]σο . . [. .]ασιν· ὅταν δὲ δέ[ῃ τὰς δημοσίας γραφ]ὰς
ε[ἰς ἁ εἰ]σαγαγεῖν, συν[έρχεται β̄ δικαστή]ρια εἰ[ς] τὴν
ἡλιαίαν. τὰ . [. . . . . .]κ̣υ̣α̣[.] . εἰς φ̄ καὶ ᾱ,
τρία [δικαστήρια]. ψῆφοι δέ εἰσιν χαλκαῖ, αὐλίσκον 2
10 ἔχουσαι ἐν τῷ μέσῳ, αἱ μὲν ἡμίσειαι τετρυπημέναι, αἱ δὲ
ἡμίσειαι πλήρεις. οἱ δὲ λαχόντες ἐπὶ τὰς ψήφους, ἐπειδὰν
εἰρημένοι ὦσιν οἱ λόγοι, παραδιδόασιν ἑκάστῳ τῶν δικαστῶν
δύο ψήφους, τετρυπημένην καὶ πλήρη, φανερὰς ὁρᾶν τοῖς
ἀντιδίκοις, ἵνα μήτε πλήρεις μήτε τετρυπημένας ἀμφοτέρας
15 λαμβάνωσι. [τότε δ' ὁ ἐπὶ τοῦτο] εἰληχὼς ἀπολα[μβάνει
τὰς βακτηρίας ἀνθ'] ὧ[ν] εἷς ἕκαστος ψηφιζ[όμενος λαμβάνει
σύμβολο]ν χαλκοῦν [μετὰ] του γ̄ (ἀποδιδοὺς γὰρ γ̄ λαμβάνει),
ἵνα ψηφίζωνται πάντες· οὐ γὰρ ἔστι λαβεῖν σύμβολον
[οὐδεν]ὶ ἐὰν μὴ ψηφίζηται. εἰσὶ δὲ ἀμφορεῖς δύο κείμενοι 3
20 ἐν τῷ δικαστηρίῳ, ὁ μὲν χαλκοῦς, ὁ δὲ ξύλινος, διαίρετοι
ὅπως μὴ λ[άθ]ῃ ὑποβάλλων [τις ψή]φους, εἰς οὓς ψηφίζονται
οἱ δικασταί, ὁ μὲν χαλκοῦς κύριος, ὁ δὲ ξύλινος ἄκυρος,
ἔχων ὁ χαλκοῦς ἐπίθημα διερρινημένον ὥστ' αὐτὴν μόνην
χωρεῖν τὴν ψῆφον, ἵνα μὴ δύο ὁ αὐτὸς [ἐμβάλλ]ῃ. ἐπειδὰν 4
25 δὲ διαψηφίζεσθαι μέλλωσιν οἱ δικασταί, ὁ κῆρυξ ἀγορεύει
πρῶτον, ἂν ἐπισκή[πτων]ται οἱ ἀντίδικοι ταῖς μαρτυρίαις.

2 ἢ θάνατος B : ἢ φυγὴ K-W ; spatium et hoc et illud capere potest 5 τὰ δὲ πολλὰ Wilcken 6-9 δέῃ . . . τρία δικαστήρια restit. K-W (praeter εἰς ἁ, quae propter spatium addidi) 9-15 ψῆφοι . . . λαμβάνωσι cit. Harp. s.v τετρυπημένη, unde lacunas supplevi 14 ἀμφοτέρας om. Harp. 15 τότε δ' ὁ ἐπὶ τοῦτο K : ὁ δὲ ταύτην τὴν ἀρχὴν Thalheim 16 τὰς βακτηρίας Photiades : τὰ σύμβολα Thalheim λαμβάνει σύμβολον B 17 μετὰ B 19 οὐδενὶ K-W 19-23 δύο . . . μόνην e Schol. Ar. Eq. 1150 restitutum 24 ἐμβάλλῃ H-L (spatio nimis, sed εν suprascriptum fuerit) 26 ἐπισκήπτωνται K-W τας μαρτυριας Ⱡ : corr. K-W e Plat. Legg. xi. 937 b

5*

Printed LXVII, Internet Archive leaf 81, 68.2–3. Edition first published 1920; scanned imprint lists reprints through 1958.

οὐ γὰρ [ἔστιν] ἐπισκήψασθαι ὅταν ἄρξωνται διαψηφίζεσθαι. ἔπειτα πάλιν ἀνακηρύττει· 'ἡ τετρυπημένη τοῦ πρότερον λέγοντος, ἡ δὲ πλήρης τοῦ ὕστερον λέγοντος.' ὁ δὲ δικαστὴς λα[βὼν ἅμα] ἐκ τοῦ λυχνείου τὰς ψήφους, πιέζων τὸν [αὐλίσκον] τῆς ψήφου καὶ οὐ δεικνύων τοῖς ἀγωνιζομένοις οὔτε τὸ τετρυπημένον οὔτε τὸ πλῆρες, ἐμβάλλει τὴν μὲν κυρίαν εἰς τὸν χαλκοῦν ἀμφορέα, τὴν δὲ ἄκυρον εἰς τὸν ξύλινον.

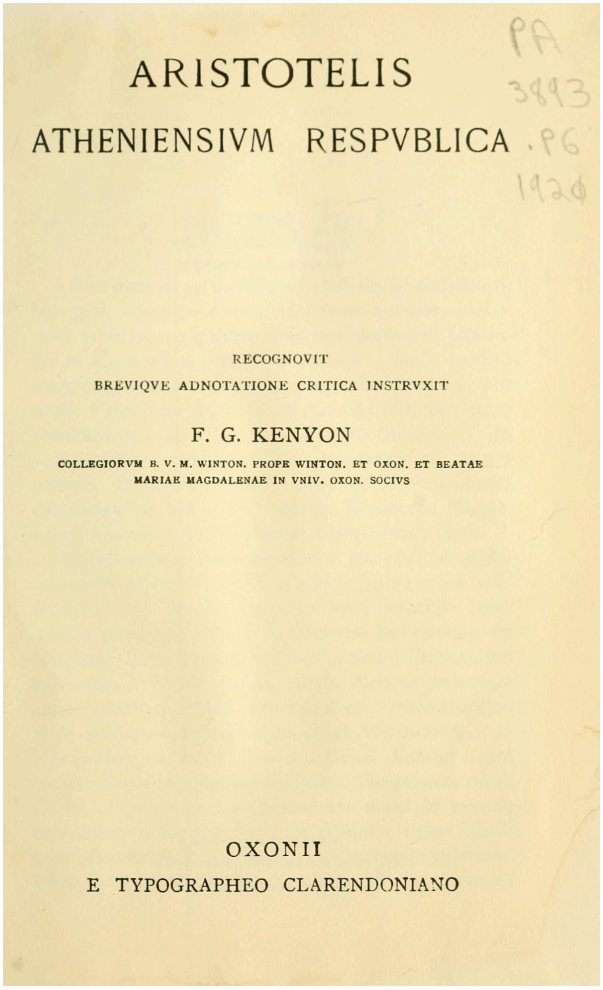
69 Πάντες δ' ἐπειδὰν ὦσι διε[ψηφισ]μένοι, λαβόντες οἱ ὑπηρέται τὸν ἀμφορέα τὸν κύριον ἐξερῶσιν ἐπὶ ἄβακα τρυπήματα ἔχοντα ὅσαιπερ εἰσὶν αἱ ψῆφοι, καὶ ταῦτα ὅ[πως] αἱ κυρ⟨ίαι προ⟩κείμεναι εὐαρίθμητοι ὦσιν, καὶ τὰ τρυπητὰ καὶ τὰ πλήρη, [[δηλονότι τοῖς ἀντιδίκοις]]. οἱ δὲ ἐπὶ τὰς ψήφους εἰληχότες δια[ριθμοῦσιν] αὐτὰς ἐπὶ τοῦ ἄβακος, [χωρὶ]ς μὲν τὰς πλήρεις, χωρὶς δὲ τὰς τετρυπημένας, καὶ ἀναγορεύει ὁ κῆρυξ τὸν [ἀριθ]μὸν τῶν ψήφων, τοῦ μὲν διώκοντος τὰς τετρυπημένας, τοῦ δὲ φεύγοντος τὰς πλήρεις· ὁποτέρῳ δ' ἂν
2 πλείων γένηται, οὗτος νικᾷ, ἂν δὲ ἴσαι, ὁ φεύγων. ἔπειτα πάλιν τιμῶσι, ἂν δέῃ τιμῆσαι, τὸν αὐτὸν τρόπον ψηφιζόμενοι, τὸ μὲν σύμβολον ἀποδιδόντες, βακτηρίαν δὲ πάλιν παραλαμβάνοντες. ἡ δὲ τίμησίς ἐστιν πρὸς ἡμίχουν ὕδατος ἑκατέρῳ. ἐπειδὰν δὲ αὐτοῖς ᾖ δεδικασμένα τὰ ἐκ τῶν νόμων, ἀπολαμβάνουσιν τὸν μισθὸν ἐν τῷ μέρει οὗ ἔλαχον ἕκαστοι.

1 ἔστιν B 4 λαβὼν ἅμα B 5 αὐλίσκον Papageorgios 9 διεψηφισμένοι Wilcken 11–12 καὶ . . . προκείμεναι K : [ἵν'] αὗται [φανεραὶ] προκείμεναι K-W ³ 13 δηλ Ł : δηλονότι Kaibel : δηλονότι τοῖς ἀντιδίκοις quasi scholium secl. Diels : δῆλα Thalheim 14–15 διαριθμοῦσιν . . . τετρυπημένας restit. B 16 ἀριθμὸν K-W 16–18 τοῦ μὲν . . . φεύγων cit. Lex. Rhet. Cantabr. (670. 30) 18 πλείους γένωνται Lex. Rhet. Cantabr., quod seq. K-W

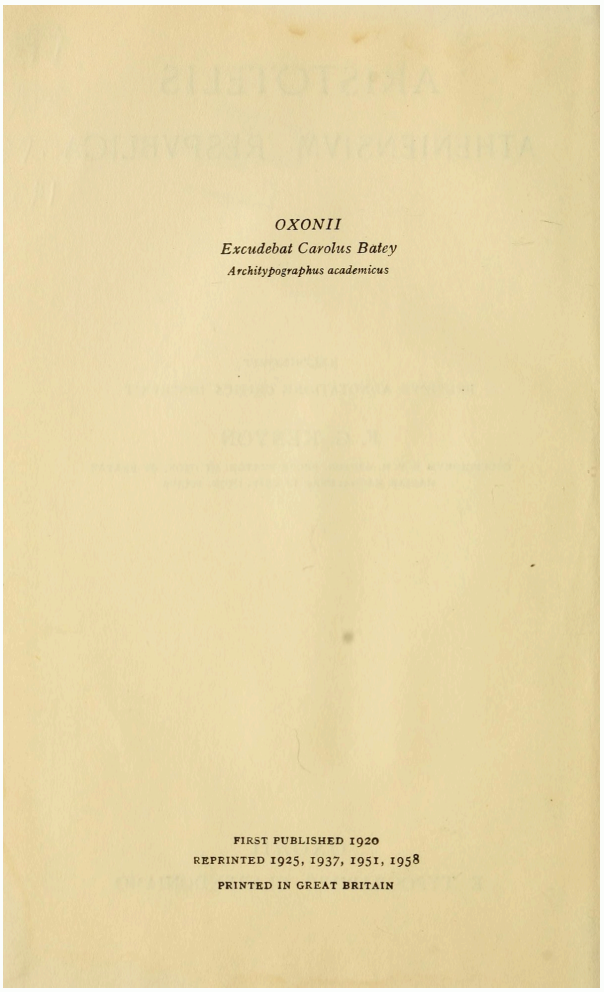
Printed LXVIII, Internet Archive leaf 82, 68.4–69.

Identify the printed volume

The title names F. G. Kenyon and Clarendon Press. The imprint says first published 1920, and lists reprints in 1925, 1937, 1951 and 1958. The edition date does not date the scanned physical copy. These complete pages preserve that distinction.



Title page, archive leaf 7.



Imprint, archive leaf 8.

Surviving objects



Zde, 2022, Museum of the Ancient Agora; objects dated around 300 BCE by the photographer. Wikimedia's rescaled 1280-pixel image. Original and attribution, CC BY-SA 4.0.

Boundaries and credits

This is a study of the procedure in Constitution of the Athenians 68.2–69.1, conventionally attributed to Aristotle, within its late fourth-century BCE setting. Authorship is debated. It is not a reconstruction of every Athenian court or an actual dispute. The training panel, pinning, simultaneous deposits, isolated observer views and counting interventions are new. The count tests a described rule, not the merits of any litigation. The work omits juror selection, evidence challenges, pay and penalty assessment. This learning app stores private choices and is not a private voting system. Ancient participation was unequal and restricted.

The Greek is transcribed from the retained print. Brackets preserve the editor's supplies or bracketed text; punctuation and opening capitalization are normalized. The new English, notes and diagrams are by Anthony Mercer.

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